

NMAP REFERENCE & CHEAT SHEET

Network Discovery, Port Scanning & Security Auditing Guide

BLUE-PURPLE EDITION

1. TARGET SPECIFICATION

Syntax / Command	Description
<code>nmap <IP></code>	Scan single IP target
<code>nmap <hostname></code>	Scan domain/hostname
<code>nmap <IP1> <IP2></code>	Scan multiple distinct targets
<code>nmap 192.168.1.1-50</code>	Scan IP address range
<code>nmap 192.168.1.0/24</code>	Scan entire CIDR subnet
<code>nmap 192.168.1.*</code>	Scan using wildcards
<code>nmap -iL targets.txt</code>	Read target list from file
<code>nmap -iR 100</code>	Choose 100 random targets
<code>nmap --exclude <IP></code>	Exclude specific hosts
<code>nmap --excludefile <file></code>	Exclude targets listed in file

2. HOST DISCOVERY (PING SCANNING)

Flag	Description	Mechanics
<code>-sn</code>	Ping Scan Only	ICMP Echo, TCP SYN/ACK (No port scan)
<code>-Pn</code>	Disable Discovery	Treats all targets as online
<code>-PE</code>	ICMP Echo Ping	Sends ICMP Echo Requests
<code>-PP</code>	ICMP Timestamp	Bypasses basic ping blocks
<code>-PM</code>	ICMP Netmask	Sends address mask queries
<code>-PS<ports></code>	TCP SYN Ping	SYN to ports (default 80); RST/ACK = UP
<code>-PA<ports></code>	TCP ACK Ping	ACK to ports; expect RST back
<code>-PU<ports></code>	UDP Ping	UDP to ports; ICMP Unreachable = UP
<code>-PR</code>	ARP Ping Scan	Default for local LAN; fast & unblockable
<code>--disable-arp-ping</code>	No ARP Ping	Forces IP-layer ping on LAN
<code>-n</code> / <code>-R</code>	DNS Resolution	<code>-n</code> Never resolve / <code>-R</code> Always resolve

3. PORT SCAN TECHNIQUES

Flag	Type	Details
<code>-sS</code>	SYN Stealth	Default (Root). SYN → SYN-ACK → RST. Stealthy.
<code>-sT</code>	Connect	Default (User). Full 3-way handshake. Logged.
<code>-sU</code>	UDP Scan	Checks UDP services. Slow, returns ICMP unreach.
<code>-sA</code>	TCP ACK	Maps firewall rulesets; returns RST on unfiltered.
<code>-sW</code>	TCP Window	Identifies open/closed via TCP Window size.
<code>-sM</code>	Maimon	Sends FIN/ACK probes. RST if closed (BSD).
<code>-sN</code>	Null Scan	Flags=0. Open drops; Closed returns RST.
<code>-sF</code>	FIN Scan	FIN bit only. Open drops; Closed returns RST.
<code>-sX</code>	Xmas Scan	FIN, URG, PSH set. Open drops; Closed returns RST.
<code>-sY</code> / <code>-sZ</code>	SCTP Scans	SCTP INIT / SCTP COOKIE ECHO scans.
<code>-sI</code>	Idle/IPID	Blind scan via zombie host IPID sequence.

6. NMAP SCRIPTING ENGINE (NSE)

Script Execution Flags:

Flag	Purpose
<code>-sC</code>	Run default safe scripts (<code>--script=default</code>)
<code>--script=<name></code>	Run single script or category (e.g. <code>vuln</code>)
<code>--script="expr"</code>	Boolean logic: "default or discovery"
<code>--script-args <a=v></code>	Pass script arguments (e.g. <code>user=admin</code>)
<code>--script-updatedb</code>	Update local script database index

Categories: `auth`, `broadcast`, `brute`, `default`, `discovery`, `dos`, `exploit`, `external`, `fuzzer`, `intrusive`, `malware`, `safe`, `vuln`.

Real-World NSE Examples:

```
# Web Server Enumeration
nmap --script http-enum,http-headers,http-methods -p 80,443 <target>

# SMB Vulnerability Check (EternalBlue)
nmap --script smb-enum-shares,smb-vuln-ms17-010 -p 445 <target>

# SSL/TLS Ciphers & Certificate
nmap --script ssl-cert,ssl-enum-ciphers -p 443 <target>

# General CVE Scanning
nmap --script vuln <target>
```

7. TIMING & OPTIMIZATION

Template	Name	Use Case
<code>-T0</code> / <code>-T1</code>	Paranoid / Sneaky	IDS Evasion (5 min / 15 sec delay)
<code>-T2</code>	Polite	Conserves bandwidth & resources
<code>-T3</code>	Normal	Default timing, dynamic latency adjustment
<code>-T4</code>	Aggressive	Fast & reliable networks (CTFs/Labs)
<code>-T5</code>	Insane	Extremely fast, may sacrifice accuracy

Performance Fine-Tuning Flags:

```
--min-rate <num> / --max-rate <num> Min/Max packets per sec
--min-parallelism / --max-parallelism Parallel probe controls
--host-timeout <time> Give up on slow target (e.g. 30m)
--max-retries <num> Limit probe retransmissions (Default 10)
```

8. FIREWALL / IDS EVASION & SPOOFING

Flag	Operational Details
<code>-f</code> / <code>-ff</code>	Fragment IP headers into 8 / 16 byte fragments
<code>--mtu <val></code>	Set custom MTU size (must be multiple of 8)
<code>-D IP1,ME,IP2</code>	Decoy scan: spoof source IP addresses
<code>-S <IP> -e <if></code>	Spoof Source IP address on specific interface
<code>-g <port></code>	Spoof source port (e.g. <code>-g 53</code> or <code>-g 80</code>)
<code>--data-length <n></code>	Append random binary payload bytes to probes
<code>--randomize-hosts</code>	Scan target IP list in non-sequential order
<code>--badsum</code>	Send invalid checksums (Firewall diagnostic)
<code>--spoof-mac <mac></code>	Spoof Ethernet MAC address (Vendor/Random)

4. PORT SELECTION & SCAN ORDER

Flag	Description
-p 22,80,443	Scan specific listed ports
-p U:53,T:80	Scan specific UDP & TCP ports
-p-	Scan all 65,535 TCP ports
-F	Fast Mode (Top 100 common ports)
--top-ports <N>	Scan top N most common ports
-r	Scan ports sequentially (not randomized)
--port-ratio <r>	Scan ports more common than ratio

5. SERVICE & OS DETECTION

Version Detection (-sV): Identifies service/app versions.

--version-intensity <0-9> (Default 7) | --version-light (2) | --version-all (9)

OS Detection (-O): TCP/UDP fingerprinting to identify OS.

--osscan-limit | --osscan-guess (Fuzzy OS match)

Aggressive Mode (-A): Combines -O , -sV , -sC , and traceroute .

9. OUTPUT OPTIONS

Flag	Format / Feature
-oN <file>	Normal human-readable terminal output
-oX <file>	XML format for automated tool integration
-oG <file>	Grepable single-line format
-oA <base>	Saves in Normal, XML, and Grepable all at once
-v / -vv	Verbose / High Verbosity mode
--reason	Displays explicit reason why a port status was set
--open	Filter output to display ONLY open ports
--packet-trace	Trace raw incoming and outgoing packets
--resume <log>	Resume aborted scan from .nmap / .gnmap log

10. OPERATIONAL COMMAND WORKFLOWS

Phase 1: Rapid Local Subnet Host Discovery

```
nmap -sn -n --min-rate 1000 192.168.1.0/24 -oG live.gnmap
```

Phase 2: All-Port Fast Surface Mapping

```
nmap -sS -p- --min-rate 2000 -Pn -n 192.168.1.50 -oN fast.nmap
```

Phase 3: Targeted Deep Service & Vuln Scanning

```
nmap -sC -sV -O -p 22,80,443,445 192.168.1.50 -oA detailed
```

Phase 4: Stealth & Firewall Evasion Scan

```
nmap -sS -Pn -f -g 53 --data-length 24 -D 192.168.1.254,ME 192.168.1.5
```